

CS464 - Data and Network Security

Hafiz Syed Ahmed Qasim
ahmed.qasim@giki.edu.pk

Chapter Overview: Finite Fields

Learning Objectives

After studying this chapter, you should be able to distinguish among groups, rings, and fields, define finite fields of the form $GF(p)$, explain the differences among ordinary polynomial arithmetic, polynomial arithmetic with coefficients in $\mathbb{Z}_p$, and modular polynomial arithmetic in $GF(2^n)$, define finite fields of the form $GF(2^n)$, and explain the two different uses of the mod operator.

Importance in Cryptography

Finite fields have become increasingly important in cryptography, and a number of cryptographic algorithms rely heavily on properties of finite fields, notably the Advanced Encryption Standard (AES) and elliptic curve cryptography, with other examples including the message authentication code CMAC and the authenticated encryption scheme GCM.

Chapter Purpose

This chapter provides the reader with sufficient background on the concepts of finite fields to be able to understand the design of AES and other cryptographic algorithms that use finite fields, approaching the topic in a way designed to enhance understanding for students unfamiliar with abstract algebra.

5.1

Groups

Introduction to groups as fundamental algebraic structures with a single binary operation.

Definition of a Group

Definition: A group G , sometimes denoted by $\{G, \cdot\}$, is a set of elements with a binary operation denoted by $\cdot$ that associates to each ordered pair (a, b) of elements in G an element $(a \cdot b)$ in G , such that the following axioms are obeyed.

The operator $\cdot$ is generic and can refer to addition, multiplication, or some other mathematical operation.

(A1) Closure

If a and b belong to G , then $a \cdot b$ is also in G .

(A2) Associative

$a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all a, b, c in G .

(A3) Identity Element

There is an element e in G such that $a \cdot e = e \cdot a = a$ for all a in G .

(A4) Inverse Element

For each a in G , there is an element a' in G such that $a \cdot a' = a' \cdot a = e$.

Group Axioms Explained

Axiom (A1) - Closure Property

The closure property states that when any two elements from the group are combined using the group operation, the result must also be an element of the same group, ensuring the operation never produces a result outside the set.

Axiom (A2) - Associative Property

The associative property ensures that when three elements are combined, the grouping of operations does not affect the result, meaning $(a \cdot b) \cdot c$ produces the same result as $a \cdot (b \cdot c)$ for all elements in the group.

Axiom (A3) - Identity Element

The identity element is a special element that, when combined with any other element using the group operation, leaves that element unchanged, analogous to how zero acts in addition or one acts in multiplication.

Axiom (A4) - Inverse Element

Every element in the group must have an inverse element such that when the element and its inverse are combined using the group operation, the result is the identity element, enabling the concept of "undoing" an operation.

Finite and Infinite Groups

Finite Groups

If a group has a finite number of elements, it is referred to as a **finite group**, and the **order of the group** is equal to the number of elements in the group.

Infinite Groups

If a group does not have a finite number of elements, the group is an **infinite group**. The set of integers under addition is an example of an infinite group.

Key Distinction: The distinction between finite and infinite groups is important because finite groups have additional structural properties that make them particularly useful in cryptography and computer science applications.

Abelian Groups

Definition of Abelian Group

A group is said to be **abelian** if it satisfies the following additional condition:

(A5) Commutative: $a \cdot b = b \cdot a$ for all a, b in G .

Example 1: Integers under Addition

The set of integers (positive, negative, and 0) under addition is an abelian group because $a + b = b + a$ for all integers a and b .

Example 2: Nonzero Real Numbers under Multiplication

The set of nonzero real numbers under multiplication is an abelian group because $a \times b = b \times a$ for all nonzero real numbers a and b .

Note: When the group operation is addition, the identity element is 0; the inverse element of a is $-a$; and subtraction is defined with the following rule: $a - b = a + (-b)$.

Cyclic Groups

Exponentiation in Groups

We define exponentiation within a group as a repeated application of the group operator, so that $a^3 = a \cdot a \cdot a$.

Furthermore, we define $a^0 = e$ as the identity element, and $a^{-n} = (a')^n$, where a' is the inverse element of a within the group.

Definition of Cyclic Group

A group G is **cyclic** if every element of G is a power a^k (k is an integer) of a fixed element $a \in G$. The element a is said to **generate** the group G or to be a **generator** of G .

Properties of Cyclic Groups

A cyclic group is always abelian and may be finite or infinite.

The additive group of integers is an infinite cyclic group generated by the element 1. In this case, powers are interpreted additively, so that n is the n th power of 1.

5.2

Rings

Introduction to rings as algebraic structures with two binary operations: addition and multiplication.

Definition of a Ring

Definition: A ring R , sometimes denoted by $\{R, +, \times\}$, is a set of elements with two binary operations, called addition and multiplication, such that for all a, b, c in R the following axioms are obeyed.

(A1–A5) Abelian Group under Addition

R is an abelian group with respect to addition; that is, R satisfies axioms A1 through A5. For the case of an additive group, we denote the identity element as 0 and the inverse of a as $-a$.

(M1) Closure under Multiplication

If a and b belong to R , then ab is also in R .

(M2) Associativity of Multiplication

$a(bc) = (ab)c$ for all a, b, c in R .

(M3) Distributive Laws

$a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ for all a, b, c in R .

Key Insight: In essence, a ring is a set of elements in which we can do addition, subtraction [$a - b = a + (-b)$], and multiplication without leaving the set.

Ring Properties and Examples

Operations in a Ring

In essence, a ring is a set of elements in which we can do addition, subtraction $[a - b = a + (-b)]$, and multiplication without leaving the set.

Division is not guaranteed in a ring, which distinguishes rings from fields.

Example: Matrix Ring

With respect to addition and multiplication, the set of all n -square matrices over the real numbers is a ring. This ring is not commutative because matrix multiplication is not commutative in general.

Example: Even Integers

Let S be the set of even integers (positive, negative, and 0) under the usual operations of addition and multiplication. S is a commutative ring because addition and multiplication of even integers always produces an even integer.

Example: $\mathbb{Z}_n$ (Integers Modulo n)

The set $\mathbb{Z}_n$ of integers $\{0, 1, \dots, n - 1\}$, together with the arithmetic operations modulo n , is a commutative ring.

Commutative Rings

Definition of Commutative Ring

A ring is said to be **commutative** if it satisfies the following additional condition:

(M4) Commutativity of Multiplication: $ab = ba$ for all a, b in R .

Example 1: Even Integers

Let S be the set of even integers (positive, negative, and 0) under the usual operations of addition and multiplication. S is a commutative ring because multiplication of even integers is commutative.

Example 2: $\mathbb{Z}_n$ Modulo n

The set $\mathbb{Z}_n$ of integers $\{0, 1, \dots, n-1\}$, together with the arithmetic operations modulo n , is a commutative ring because modular multiplication is commutative.

Non-Example: n -Square Matrices

The set of all n -square matrices defined in the preceding example is **not a commutative ring** because matrix multiplication is not commutative in general.

Integral Domains

Definition of Integral Domain

An integral domain is a commutative ring that obeys the following axioms:

(M5) Multiplicative Identity:

There is an element 1 in R such that $a \cdot 1 = 1 \cdot a = a$ for all a in R .

(M6) No Zero Divisors:

If a, b in R and $ab = 0$, then either $a = 0$ or $b = 0$.

Example: The Integers

Let S be the set of integers (positive, negative, and 0) under the usual operations of addition and multiplication. S is an integral domain because:

- It is a commutative ring (satisfies A1-A5 and M1-M4)
- It has multiplicative identity 1 (satisfies M5)
- It has no zero divisors: if $ab = 0$, then $a = 0$ or $b = 0$ (satisfies M6)

Key Property: The absence of zero divisors (M6) is crucial for many algebraic operations and ensures that the cancellation property holds: if $ab = ac$ and $a \neq 0$, then $b = c$.

5.3

Fields

Introduction to fields as the most complete algebraic structure supporting all four arithmetic operations.

Definition of a Field

Definition: A field F , sometimes denoted by $\{F, +, \times\}$, is a set of elements with two binary operations, called addition and multiplication, such that for all a, b, c in F the following axioms are obeyed:

(A1–M6) F is an integral domain; that is, F satisfies axioms A1 through A5 and M1 through M6.

(M7) Multiplicative Inverse

For each a in F , except 0, there is an element a^{-1} in F such that $aa^{-1} = (a^{-1})a = 1$.

Key Insight

In essence, a field is a set of elements in which we can do addition, subtraction, multiplication, and division without leaving the set. Division is defined with the following rule: $a/b = a(b^{-1})$.

Progression: Every field is an integral domain, every integral domain is a commutative ring, every commutative ring is a ring, and every ring is an abelian group under addition.

Field Examples and Non-Examples

Examples of Fields

1. **Rational Numbers ($\mathbb{Q}$):** The set of all rational numbers (fractions a/b where a, b are integers and $b \neq 0$) forms a field under standard addition and multiplication.
2. **Real Numbers ($\mathbb{R}$):** The set of all real numbers forms a field under standard arithmetic operations.
3. **Complex Numbers ($\mathbb{C}$):** The set of all complex numbers forms a field under complex addition and multiplication.

Non-Example: The Integers ($\mathbb{Z}$)

Note that the set of all integers is **not a field**, because not every element of the set has a multiplicative inverse. In fact, only the elements 1 and -1 have multiplicative inverses in the integers. For example, there is no integer x such that $2x = 1$.

Key Distinction: The presence of multiplicative inverses for all nonzero elements (axiom M7) is what distinguishes fields from integral domains and makes division possible.

Alternative Characterization of Fields

In gaining insight into fields, the following alternate characterization may be useful. A field F , denoted by $\{F, +, \times\}$, is a set of elements with two binary operations, called addition and multiplication, such that the following conditions hold:

Condition 1

F forms an abelian group with respect to addition. This means all elements satisfy axioms A1 through A5 with the addition operation.

Condition 2

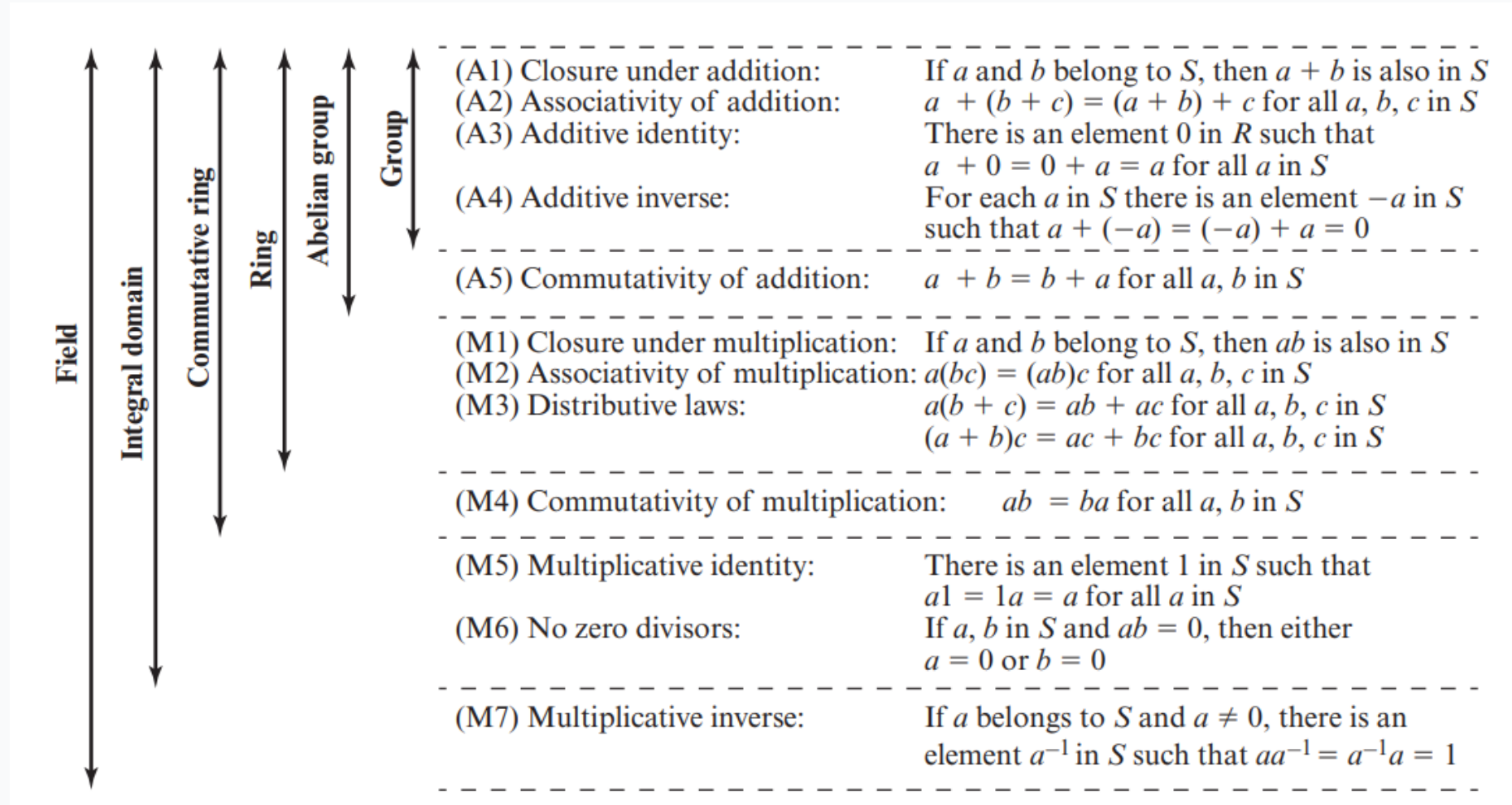
The **nonzero elements** of F form an abelian group with respect to multiplication. This means all nonzero elements satisfy axioms M1, M2, M4, M5, and M7.

Condition 3

The distributive law holds. That is, for all a, b, c in F : $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

Advantage: This characterization emphasizes that a field consists of two interacting group structures tied together by the distributive law.

Figure 5.2: Properties of Groups, Rings, and Fields

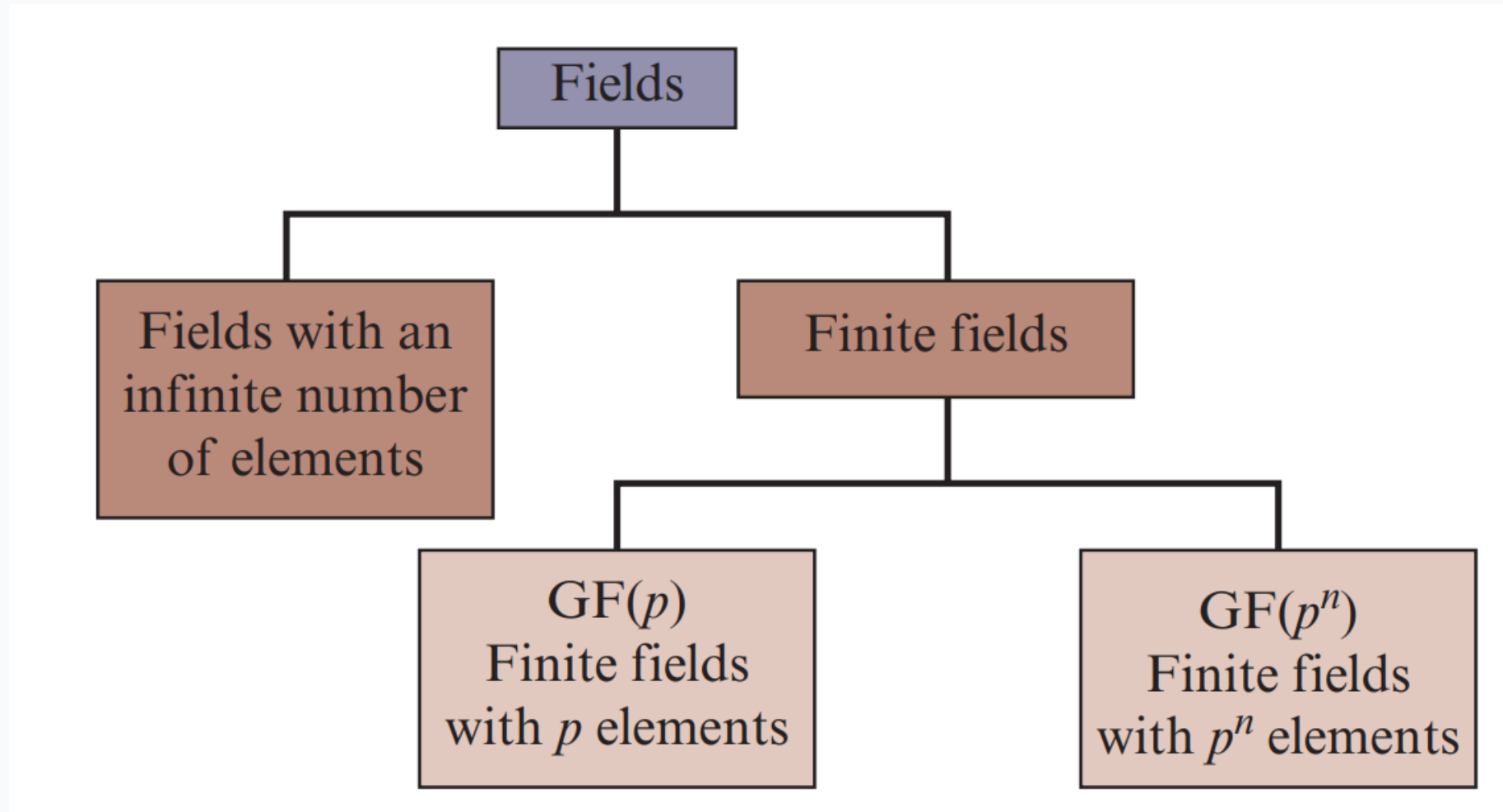


5.4

Finite Fields of the Form $\text{GF}(p)$

Introduction to finite fields with prime order and their construction using modular arithmetic.

Figure 5.3: Types of Fields



Finite Field Order Theorem

Fundamental Theorem

It can be shown that the **order of a finite field** (number of elements in the field) must be a power of a prime p^n , where n is a positive integer.

Notation: $GF(p^n)$

The finite field of order p^n is generally written $GF(p^n)$; GF stands for **Galois field**, in honor of the mathematician Galois who first studied finite fields.

Special Case 1: $GF(p)$

For $n = 1$, we have the finite field $GF(p)$; this finite field has a different structure than that for finite fields with $n > 1$ and is studied in this section.

Special Case 2: $GF(2^n)$

For finite fields of the form $GF(p^n)$, $GF(2^n)$ fields are of particular cryptographic interest, and these are covered in Section 5.6.

Definition of GF(p)

Construction of GF(p)

For a given prime, p , we define the finite field of order p , $GF(p)$, as the set Z_p of integers $\{0, 1, \dots, p - 1\}$ together with the arithmetic operations modulo p .

Key Observation

Note therefore that we are using ordinary modular arithmetic to define the operations over these fields. This is a straightforward application of the modular arithmetic.

Examples

- $GF(2)$: The set $\{0, 1\}$ with arithmetic modulo 2
- $GF(7)$: The set $\{0, 1, 2, 3, 4, 5, 6\}$ with arithmetic modulo 7
- $GF(251)$: Used in some cryptographic applications

Important: The modulus p must be prime for Z_p to be a field. If n is not prime, Z_n is only a commutative ring, not a field.

Why $\mathbb{Z}_p$ Forms a Field

Background: $\mathbb{Z}_n$ as a Commutative Ring

Recall that we showed in Section 5.2 that the set $\mathbb{Z}_n$ of integers $\{0, 1, \dots, n - 1\}$, together with the arithmetic operations modulo n , is a commutative ring.

Multiplicative Inverses in $\mathbb{Z}_n$

We further observed that any integer in $\mathbb{Z}_n$ has a multiplicative inverse if and only if that integer is relatively prime to n . If n is prime, then all of the nonzero integers in $\mathbb{Z}_n$ are relatively prime to n , and therefore there exists a multiplicative inverse for all of the nonzero integers in $\mathbb{Z}_n$.

The Key Proof

Because w is relatively prime to p , if we multiply all the elements of $\mathbb{Z}_p$ by w , the resulting residues are all of the elements of $\mathbb{Z}_p$ permuted. Thus, exactly one of the residues has the value 1. Therefore, there is some integer in $\mathbb{Z}_p$ that, when multiplied by w , yields the residue 1. That integer is the multiplicative inverse of w , designated w^{-1} .

Conclusion: Therefore, $\mathbb{Z}_p$ is in fact a finite field.

GF(2): The Simplest Finite Field

The simplest finite field is GF(2). Its arithmetic operations are easily summarized:

+	0	1
0	0	1
1	1	0

Addition

$\times$	0	1
0	0	0
1	0	1

Multiplication

w	$-w$	w^{-1}
0	0	—
1	1	1

Inverses

In this case, addition is equivalent to the exclusive-OR (XOR) operation, and multiplication is equivalent to the logical AND operation.

Important Connection

In this case, addition is equivalent to the **exclusive-OR (XOR)** operation, and multiplication is equivalent to the **logical AND** operation. This connection makes GF(2) particularly useful in computer science and digital logic.

Finding Multiplicative Inverse in GF(p)

Small Values of p

It is easy to find the multiplicative inverse of an element in GF(p) for small values of p. You simply construct a multiplication table, such as shown in Table 5.1e, and the desired result can be read directly.

Large Values of p

However, for large values of p, this approach is not practical. We need a more efficient algorithm.

Theoretical Foundation

If a and b are relatively prime, then b has a multiplicative inverse modulo a. That is, if $\gcd(a, b) = 1$, then b has a multiplicative inverse modulo a. Thus, for positive integer $b < a$, there exists a $b^{-1} < a$ such that $bb^{-1} \equiv 1 \pmod{a}$.

Application to GF(p)

If a is a prime number and $0 < b < a$, then clearly a and b are relatively prime and have a greatest common divisor of 1. We can easily compute b^{-1} using the extended Euclidean algorithm.

Extended Euclidean Algorithm for Inverses

The Algorithm

We repeat here Equation (2.7), which we showed can be solved with the extended Euclidean algorithm:

$$ax + by = d = \gcd(a, b)$$

Finding the Inverse

Now, if $\gcd(a, b) = 1$, then we have $ax + by = 1$. Using the basic equalities of modular arithmetic, we can say:

$$[(ax \bmod a) + (by \bmod a)] \bmod a = 1 \bmod a$$

$$0 + (by \bmod a) = 1$$

But if $by \bmod a = 1$, then $y = b^{-1}$. Thus, applying the extended Euclidean algorithm yields the value of the multiplicative inverse of b if $\gcd(a, b) = 1$.

Example

Consider $a = 1759$ (prime) and $b = 550$. The solution yields $y = 355$. Thus, $b^{-1} = 355$. To verify: $550 \times 355 \bmod 1759 = 195250 \bmod 1759 = 1$.

Table 5.1: Arithmetic Modulo 8 and Modulo 7

+	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

(a) Addition modulo 8

+	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	0
2	2	3	4	5	6	0	1
3	3	4	5	6	0	1	2
4	4	5	6	0	1	2	3
5	5	6	0	1	2	3	4
6	6	0	1	2	3	4	5

(d) Addition modulo 7

$\times$	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	0	2	4	6
3	0	3	6	1	4	7	2	5
4	0	4	0	4	0	4	0	4
5	0	5	2	7	4	1	6	3
6	0	6	4	2	0	6	4	2
7	0	7	6	5	4	3	2	1

(b) Multiplication modulo 8

$\times$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	1	3	5
3	0	3	6	2	5	1	4
4	0	4	1	5	2	6	3
5	0	5	3	1	6	4	2
6	0	6	5	4	3	2	1

(e) Multiplication modulo 7

w	0	1	2	3	4	5	6	7
$-w$	0	7	6	5	4	3	2	1
w^{-1}	—	1	—	3	—	5	—	7

(c) Additive and multiplicative inverses modulo 8

w	0	1	2	3	4	5	6
$-w$	0	6	5	4	3	2	1
w^{-1}	—	1	4	5	2	3	6

(f) Additive and multiplicative inverses modulo 7

Summary of GF(p)

In this section, we have shown how to construct a finite field of order p , where p is prime. Specifically, we defined $GF(p)$ with the following properties:

Property 1

$GF(p)$ consists of p elements: $\{0, 1, \dots, p-1\}$.

Property 2

The binary operations $+$ and $\times$ are defined over the set. The operations of addition, subtraction, multiplication, and division can be performed without leaving the set. Each element of the set other than 0 has a multiplicative inverse, and division is performed by multiplication by the multiplicative inverse.

Key Characteristics

The elements of $GF(p)$ are the integers $\{0, 1, \dots, p-1\}$ and the arithmetic operations are addition and multiplication mod p .

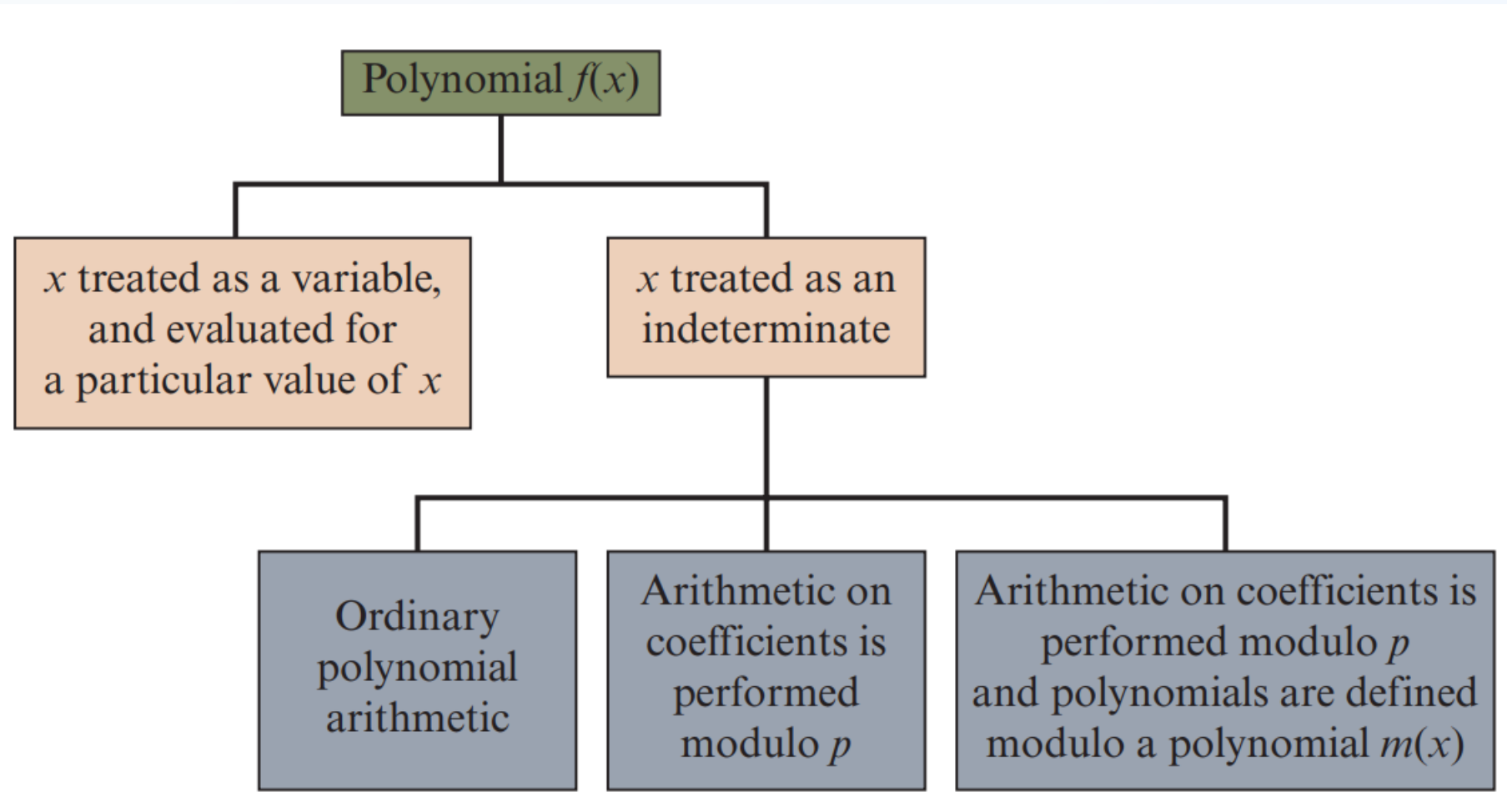
Significance: $GF(p)$ fields are used in certain asymmetric cryptographic algorithms, providing the mathematical foundation for operations requiring division.

5.5

Polynomial Arithmetic

Introduction to polynomial arithmetic as the foundation for constructing $\text{GF}(2^n)$ fields.

Figure 5.4: Treatment of Polynomials



Ordinary Polynomial Arithmetic

Definition of a Polynomial

A polynomial of degree n (integer $n \geq 0$) is an expression of the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = \sum_{i=0}^n a_i x^i$$

where the a_i are elements of some designated set of numbers S , called the coefficient set, and $a_n \neq 0$. We say that such polynomials are defined over the coefficient set S .

Special Cases

- **Constant polynomial:** A zero-degree polynomial is simply an element of the set of coefficients.
- **Monic polynomial:** An n th-degree polynomial is monic if $a_n = 1$.

The Indeterminate

In the context of abstract algebra, we are usually not interested in evaluating a polynomial for a particular value of x . To emphasize this point, the variable x is sometimes referred to as the **indeterminate**.

Operations: Polynomial arithmetic includes the operations of addition, subtraction, multiplication, and division. These operations are defined in a natural way as though the variable x was an element of S . Division requires that S be a field.

Polynomial Addition and Multiplication

Addition and Subtraction

Addition and subtraction are performed by adding or subtracting corresponding coefficients. Thus, if:

$$f(x) = \sum_{i=0}^n a_i x^i \text{ and } g(x) = \sum_{i=0}^m b_i x^i \text{ where } n \geq m$$

Then addition is defined as: $f(x) + g(x) = \sum_{i=0}^m (a_i + b_i) x^i + \sum_{i=m+1}^n a_i x^i$

Multiplication

Multiplication is defined as: $f(x) \times g(x) = \sum_{i=0}^{n+m} c_i x^i$ where $c_k = a_0 b_k + a_1 b_{k-1} + \dots + a_{k-1} b_1 + a_k b_0$

Note that the degree of the product is equal to the sum of the degrees of the two polynomials.

Example

Let $f(x) = x^3 + x^2 + 2$ and $g(x) = x^2 - x + 1$, where S is the set of integers. Then:

- $f(x) + g(x) = x^3 + 2x^2 - x + 3$
- $f(x) - g(x) = x^3 + x + 1$
- $f(x) \times g(x) = x^5 + 3x^2 - 2x + 2$

Figure 5.5: Examples of Polynomial Arithmetic

$$\begin{array}{r} x^3 + x^2 \quad + 2 \\ + \quad (x^2 - x + 1) \\ \hline x^3 + 2x^2 - x + 3 \end{array}$$

(a) Addition

$$\begin{array}{r} x^3 + x^2 \quad + 2 \\ - \quad (x^2 - x + 1) \\ \hline x^3 \quad + x + 1 \end{array}$$

(b) Subtraction

$$\begin{array}{r} x^3 + x^2 \quad + 2 \\ \times \quad (x^2 - x + 1) \\ \hline x^3 + x^2 \quad + 2 \\ -x^4 - x^3 \quad - 2x \\ \hline x^5 + x^4 \quad + 2x^2 \\ \hline x^5 \quad + 3x^2 - 2x + 2 \end{array}$$

(c) Multiplication

$$\begin{array}{r} x^2 - x + 1 \overline{) x^3 + x^2 + 2} \\ \underline{x^3 - x^2 + x} \\ 2x^2 - x + 2 \\ \underline{2x^2 - 2x + 2} \\ x \end{array}$$

(d) Division

Polynomial Arithmetic with Coefficients in $\mathbb{Z}_p$

Polynomials Over a Field

Let us now consider polynomials in which the coefficients are elements of some field F ; we refer to this as a polynomial over the field F . In this case, it is easy to show that the set of such polynomials is a ring, referred to as a polynomial ring.

Division in Polynomial Rings

When polynomial arithmetic is performed on polynomials over a field, then division is possible. Note that this does not mean that exact division is possible.

Distinction: Within a field, given two elements a and b , the quotient a/b is also an element of the field. However, given a ring R that is not a field, in general, division will result in both a quotient and a remainder; this is not exact division.

Rational Numbers (Field)

$5/3 = 5/3$ (exact, in the field)

$\text{GF}(7)$ (Field)

$5/3 = (5 \times 3^{-1}) \bmod 7 = (5 \times 5) \bmod 7 = 4$ (exact)

Integers (Ring, not Field)

$5/3 = 1 + 2/3$ (not exact, has remainder)

Division Algorithm for Polynomials

The Division Algorithm

We can restate the division algorithm for polynomials over a field as follows. Given polynomials $f(x)$ of degree n and $g(x)$ of degree m ($n \geq m$), if we divide $f(x)$ by $g(x)$, we get a quotient $q(x)$ and a remainder $r(x)$ that obey the relationship:

$$f(x) = q(x)g(x) + r(x)$$

Polynomial Degrees

- Degree $f(x) = n$
- Degree $g(x) = m$
- Degree $q(x) = n - m$
- $0 \leq \text{Degree } r(x) \leq m - 1$

Modulo Operation and Divisibility

We can write $f(x) \bmod g(x)$ for the remainder $r(x)$. If there is no remainder [i.e., $r(x) = 0$], then we can say $g(x)$ divides $f(x)$, written as $g(x) \mid f(x)$. Equivalently, we can say that $g(x)$ is a factor of $f(x)$ or $g(x)$ is a divisor of $f(x)$.

Polynomial Arithmetic over GF(2)

GF(2) Arithmetic Properties

For our purposes, polynomials over GF(2) are of most interest. GF(2):

Addition = XOR

$$1 + 1 = 0$$

$$1 + 0 = 1$$

$$0 + 1 = 1$$

$$0 + 0 = 0$$

Multiplication = AND

$$1 \times 1 = 1$$

$$1 \times 0 = 0$$

$$0 \times 1 = 0$$

$$0 \times 0 = 0$$

Furthermore, addition and subtraction are equivalent mod 2: $1 + 1 = 1 - 1 = 0$, $1 + 0 = 1 - 0 = 1$, $0 + 1 = 0 - 1 = 1$.

Example Polynomials

For $f(x) = (x^7 + x^5 + x^4 + x^3 + x + 1)$ and $g(x) = (x^3 + x + 1)$, Figure 5.6 shows $f(x) + g(x)$, $f(x) - g(x)$, $f(x) \times g(x)$, and $f(x)/g(x)$. Note that $g(x)$ divides $f(x)$.

Significance: GF(2) polynomial arithmetic is fundamental to the construction of GF(2ⁿ) fields used in AES and other cryptographic algorithms.

Figure 5.6: Polynomial Arithmetic over GF(2)

$$\begin{array}{r} x^7 \quad +x^5+x^4+x^3 \quad +x+1 \\ + (x^3 \quad +x+1) \\ \hline x^7 \quad +x^5+x^4 \end{array}$$

(a) Addition

$$\begin{array}{r} x^7 \quad +x^5+x^4+x^3 \quad +x+1 \\ - (x^3 \quad +x+1) \\ \hline x^7 \quad +x^5+x^4 \end{array}$$

(b) Subtraction

$$\begin{array}{r} x^7 \quad +x^5+x^4+x^3 \quad +x+1 \\ \times (x^3 \quad +x+1) \\ \hline x^7 \quad +x^5+x^4+x^3 \quad +x+1 \\ x^8 \quad +x^6+x^5+x^4 \quad +x^2+x \\ x^{10} \quad +x^8+x^7+x^6 \quad +x^4+x^3 \\ \hline x^{10} \quad \quad \quad +x^4 \quad +x^2 \quad +1 \end{array}$$

(c) Multiplication

$$\begin{array}{r} x^4+1 \\ x^3+x+1 \overline{) x^7+x^5+x^4+x^3+x+1} \\ \underline{x^7+x^5+x^4} \\ x^3+x+1 \\ \underline{x^3+x+1} \\ \end{array}$$

(d) Division

Irreducible Polynomials

Definition

A polynomial $f(x)$ over a field F is called **irreducible** if and only if $f(x)$ cannot be expressed as a product of two polynomials, both over F , and both of degree greater than 0 and lower than that of $f(x)$. By analogy to integers, an irreducible polynomial is also called a **prime polynomial**.

Example: Reducible Polynomial

The polynomial $f(x) = x^4 + 1$ over $\text{GF}(2)$ is reducible, because:

$$x^4 + 1 = (x + 1)(x^3 + x^2 + x + 1)$$

Example: Irreducible Polynomial

Consider the polynomial $f(x) = x^3 + x + 1$. It is clear by inspection that x is not a factor of $f(x)$. We easily show that $x + 1$ is not a factor of $f(x)$ by polynomial division.

Thus, $f(x)$ has no factors of degree 1. But it is clear by inspection that if $f(x)$ is reducible, it must have one factor of degree 2 and one factor of degree 1. Therefore, $f(x)$ is irreducible.

Greatest Common Divisor of Polynomials

Extending the Analogy

We can extend the analogy between polynomial arithmetic over a field and integer arithmetic by defining the greatest common divisor as follows.

Definition of GCD for Polynomials

The polynomial $c(x)$ is said to be the greatest common divisor of $a(x)$ and $b(x)$ if the following are true:

1. $c(x)$ divides both $a(x)$ and $b(x)$.
2. Any divisor of $a(x)$ and $b(x)$ is a divisor of $c(x)$.

Equivalent Definition

An equivalent definition is the following: $\gcd[a(x), b(x)]$ is the polynomial of maximum degree that divides both $a(x)$ and $b(x)$.

Euclidean Algorithm Adaptation

We can adapt the Euclidean algorithm to compute the greatest common divisor of two polynomials. Recall that $\gcd(a, b) = \gcd(b, a \bmod b)$ for integers. This equality can be rewritten as:

$$\gcd[a(x), b(x)] = \gcd[b(x), a(x) \bmod b(x)]$$

Euclidean Algorithm for Polynomials - Example

Problem

Find $\gcd[a(x), b(x)]$ for $a(x) = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$ and $b(x) = x^4 + x^2 + x + 1$.

Step 1: Divide $a(x)$ by $b(x)$

First, we divide $a(x)$ by $b(x)$ using polynomial long division over $\text{GF}(2)$. This yields:

- Quotient $q_1(x) = x^2 + x$
- Remainder $r_1(x) = x^3 + x^2 + 1$

Step 2: Divide $b(x)$ by $r_1(x)$

Then, we divide $b(x)$ by $r_1(x)$. This yields:

- Quotient $q_2(x) = x + 1$
- Remainder $r_2(x) = 0$

Result

Therefore, $\gcd[a(x), b(x)] = r_1(x) = x^3 + x^2 + 1$.

Summary of Polynomial Arithmetic

We began this section with a discussion of arithmetic with ordinary polynomials. Here is a summary of what we covered:

Ordinary Polynomial Arithmetic

In ordinary polynomial arithmetic, the variable is not evaluated; that is, we do not plug a value in for the variable of the polynomials. Instead, arithmetic operations are performed on polynomials (addition, subtraction, multiplication, division) using the ordinary rules of algebra. Polynomial division is not allowed unless the coefficients are elements of a field.

Polynomial Arithmetic over $\text{GF}(p)$

Next, we discussed polynomial arithmetic in which the coefficients are elements of $\text{GF}(p)$. In this case, polynomial addition, subtraction, multiplication, and division are allowed. However, division is not exact; that is, in general division results in a quotient and a remainder.

Euclidean Algorithm for Polynomials

Finally, we showed that the Euclidean algorithm can be extended to find the greatest common divisor of two polynomials whose coefficients are elements of a field.

Foundation: All of the material in this section provides a foundation for the following section, in which polynomials are used to define finite fields of order p^n .

5.6

Finite Fields of the Form $\text{GF}(2^n)$

Introduction to finite fields used in AES and other cryptographic algorithms.

Motivation for $\text{GF}(2^n)$

The Problem with Encryption Algorithms

Virtually all encryption algorithms, both symmetric and asymmetric, involve arithmetic operations on integers. If one of the operations that is used in the algorithm is division, then we need to work in arithmetic defined over a field.

Efficiency Requirements

For convenience and for implementation efficiency, we would also like to work with integers that fit exactly into a given number of bits with no wasted bit patterns. That is, we wish to work with integers in the range 0 through $2^n - 1$, which fit into an n -bit word.

Example: 8-bit Operations

Suppose we wish to define a conventional encryption algorithm that operates on data 8 bits at a time, and we wish to perform division. With 8 bits, we can represent integers in the range 0 through 255. However, 256 is not a prime number, so that if arithmetic is performed in $\mathbb{Z}_{256}$ (arithmetic modulo 256), this set of integers will not be a field.

The Prime Alternative

The closest prime number less than 256 is 251. Thus, the set $\mathbb{Z}_{251}$, using arithmetic modulo 251, is a field. However, in this case the 8-bit patterns representing the integers 251 through 255 would not be used, resulting in inefficient use of storage.

Comparison: Z_8 vs $GF(2^3)$

The Problem with Z_{2^n}

As the preceding example points out, if all arithmetic operations are to be used and we wish to represent a full range of integers in n bits, then arithmetic modulo 2^n will not work. Equivalently, the set of integers modulo 2^n for $n > 1$, is not a field.

Example with 3-bit Blocks

Suppose we wish to use 3-bit blocks in our encryption algorithm and use only the operations of addition and multiplication. Then arithmetic modulo 8 is well defined. However, note that in the multiplication table, the nonzero integers do not appear an equal number of times.

Observation: The uneven distribution in Z_8 (especially 12 occurrences of 4) could be cryptographically weaker than the uniform distribution in $GF(2^3)$.

Occurrence Counts Comparison

Integer	Occurrences in Z_8	Occurrences in $GF(2^3)$
1	4	7
2	8	7
3	4	7
4	12	7
5	4	7
6	8	7
7	4	7

Table 5.2: Arithmetic in GF(2³)

		000	001	010	011	100	101	110	111
+		0	1	2	3	4	5	6	7
000	0	0	1	2	3	4	5	6	7
001	1	1	0	3	2	5	4	7	6
010	2	2	3	0	1	6	7	4	5
011	3	3	2	1	0	7	6	5	4
100	4	4	5	6	7	0	1	2	3
101	5	5	4	7	6	1	0	3	2
110	6	6	7	4	5	2	3	0	1
111	7	7	6	5	4	3	2	1	0

(a) Addition

		000	001	010	011	100	101	110	111
×		0	1	2	3	4	5	6	7
000	0	0	0	0	0	0	0	0	0
001	1	0	1	2	3	4	5	6	7
010	2	0	2	4	6	3	1	7	5
011	3	0	3	6	5	7	4	1	2
100	4	0	4	3	7	6	2	5	1
101	5	0	5	1	4	2	7	3	6
110	6	0	6	7	1	5	3	2	4
111	7	0	7	5	2	1	6	4	3

(b) Multiplication

w	-w	w ⁻¹
0	0	—
1	1	1
2	2	5
3	3	6
4	4	7
5	5	2
6	6	3
7	7	4

(c) Additive and multiplicative inverses

Observations about $\text{GF}(2^3)$

For the moment, let us set aside the question of how the matrices of Table 5.2 were constructed and instead make some observations:

Observation 1

The addition and multiplication tables are symmetric about the main diagonal, in conformance to the commutative property of addition and multiplication. This property is also exhibited in Table 5.1, which uses mod 8 arithmetic.

Observation 2

All the nonzero elements defined by Table 5.2 have a multiplicative inverse, unlike the case with Table 5.1.

Observation 3

The scheme defined by Table 5.2 satisfies all the requirements for a finite field. Thus, we can refer to this scheme as $\text{GF}(2^3)$.

Observation 4

For convenience, we show the 3-bit assignment used for each of the elements of $\text{GF}(2^3)$.

Modular Polynomial Arithmetic

The Set S of Polynomials

Consider the set S of all polynomials of degree $n - 1$ or less over the field $\mathbb{Z}_p$. Thus, each polynomial has the form:

$$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0 = \sum_{i=0}^{n-1} a_i x^i$$

where each a_i takes on a value in the set $\{0, 1, \dots, p - 1\}$. There are a total of p^n different polynomials in S.

Three Key Elements of the Definition

1. Arithmetic follows the ordinary rules of polynomial arithmetic using the basic rules of algebra, with the following two refinements.
2. Arithmetic on the coefficients is performed modulo p . That is, we use the rules of arithmetic for the finite field $\mathbb{Z}_p$.
3. If multiplication results in a polynomial of degree greater than $n - 1$, then the polynomial is reduced modulo some irreducible polynomial $m(x)$ of degree n . That is, we divide by $m(x)$ and keep the remainder. For a polynomial $f(x)$, the remainder is expressed as $r(x) = f(x) \bmod m(x)$.

Example Polynomials in S

Example 1: $p = 3, n = 2$

For $p = 3$ and $n = 2$, the $3^2 = 9$ polynomials in the set are:

$0, 1, 2, x, x + 1, x + 2, 2x, 2x + 1, 2x + 2$

Example 2: $p = 2, n = 3$

For $p = 2$ and $n = 3$, the $2^3 = 8$ polynomials in the set are:

$0, 1, x, x + 1, x^2, x^2 + 1, x^2 + x, x^2 + x + 1$

Key Point: With the appropriate definition of arithmetic operations, each such set S is a finite field. The definition consists of the three elements described in the previous slide.

AES Example: Multiplication in $GF(2^8)$

The Advanced Encryption Standard (AES) uses arithmetic in the finite field $\text{GF}(2^8)$, with the irreducible polynomial $m(x) = x^8 + x^4 + x^3 + x + 1$. Consider the two polynomials $f(x) = x^6 + x^4 + x^2 + x + 1$ and $g(x) = x^7 + x + 1$. Then

$$\begin{aligned} f(x) + g(x) &= x^6 + x^4 + x^2 + x + 1 + x^7 + x + 1 \\ &= x^7 + x^6 + x^4 + x^2 \end{aligned}$$

$$\begin{aligned} f(x) \times g(x) &= x^{13} + x^{11} + x^9 + x^8 + x^7 \\ &\quad + x^7 + x^5 + x^3 + x^2 + x \\ &\quad + x^6 + x^4 + x^2 + x + 1 \\ &= x^{13} + x^{11} + x^9 + x^8 + x^6 + x^5 + x^4 + x^3 + 1 \end{aligned}$$

$$\begin{array}{r} x^8 + x^4 + x^3 + x + 1 \sqrt{x^{13} + x^{11} + x^9 + x^8} \qquad + x^6 + x^5 + x^4 + x^3 + 1 \\ \underline{x^{13} \qquad \qquad + x^9 + x^8 \qquad \qquad + x^6 + x^5} \\ \qquad \qquad \qquad x^{11} \qquad \qquad \qquad + x^4 + x^3 \\ \qquad \qquad \qquad \underline{x^{11} \qquad \qquad + x^7 + x^6 \qquad \qquad + x^4 + x^3} \\ \qquad \qquad \qquad \qquad \qquad \qquad x^7 + x^6 \qquad \qquad \qquad + 1 \end{array}$$

Therefore, $f(x) \times g(x) \bmod m(x) = x^7 + x^6 + 1$.

Constructing $\text{GF}(2^3)$

Choosing an Irreducible Polynomial

To construct the finite field $\text{GF}(2^3)$, we need to choose an irreducible polynomial of degree 3. There are only two such polynomials:

$$(x^3 + x^2 + 1) \text{ and } (x^3 + x + 1)$$

Using $x^3 + x + 1$

Using the latter polynomial $(x^3 + x + 1)$, Table 5.3 shows the addition and multiplication tables for $\text{GF}(2^3)$. Note that this set of tables has the identical structure to those of Table 5.2. Thus, we have succeeded in finding a way to define a field of order 2^3 .

Reading the Tables

We can now read additions and multiplications from the table easily. For example:

- Consider binary $100 + 010 = 110$. This is equivalent to $x^2 + x$.
- Also consider $100 \times 010 = 011$, which is equivalent to $x^2 \times x = x^3$ and reduces to $x + 1$. That is, $x^3 \bmod (x^3 + x + 1) = x + 1$, which is equivalent to 011.

Table 5.3: GF(2³) with Polynomial Representation

		000	001	010	011	100	101	110	111
	+	0	1	x	$x + 1$	x^2	$x^2 + 1$	$x^2 + x$	$x^2 + x + 1$
000	0	0	1	x	$x + 1$	x^2	$x^2 + 1$	$x^2 + x$	$x^2 + x + 1$
001	1	1	0	$x + 1$	x	$x^2 + 1$	x^2	$x^2 + x + 1$	$x^2 + x$
010	x	x	$x + 1$	0	1	$x^2 + x$	$x^2 + x + 1$	x^2	$x^2 + 1$
011	$x + 1$	$x + 1$	x	1	0	$x^2 + x + 1$	$x^2 + x$	$x^2 + 1$	x^2
100	x^2	x^2	$x^2 + 1$	$x^2 + x$	$x^2 + x + 1$	0	1	x	$x + 1$
101	$x^2 + 1$	$x^2 + 1$	x^2	$x^2 + x + 1$	$x^2 + x$	1	0	$x + 1$	x
110	$x^2 + x$	$x^2 + x$	$x^2 + x + 1$	x^2	$x^2 + 1$	x	$x + 1$	0	1
111	$x^2 + x + 1$	$x^2 + x + 1$	$x^2 + x$	$x^2 + 1$	x^2	$x + 1$	x	1	0

(a) Addition

		000	001	010	011	100	101	110	111
	$\times$	0	1	x	$x + 1$	x^2	$x^2 + 1$	$x^2 + x$	$x^2 + x + 1$
000	0	0	0	0	0	0	0	0	0
001	1	0	1	x	$x + 1$	x^2	$x^2 + 1$	$x^2 + x$	$x^2 + x + 1$
010	x	0	x	x^2	$x^2 + x$	$x + 1$	1	$x^2 + x + 1$	$x^2 + 1$
011	$x + 1$	0	$x + 1$	$x^2 + x$	$x^2 + 1$	$x^2 + x + 1$	x^2	1	x
100	x^2	0	x^2	$x + 1$	$x^2 + x + 1$	$x^2 + x$	x	$x^2 + 1$	1
101	$x^2 + 1$	0	$x^2 + 1$	1	x^2	x	$x^2 + x + 1$	$x + 1$	$x^2 + x$
110	$x^2 + x$	0	$x^2 + x$	$x^2 + x + 1$	1	$x^2 + 1$	$x + 1$	x	x^2
111	$x^2 + x + 1$	0	$x^2 + x + 1$	$x^2 + 1$	x	1	$x^2 + 1$	x^2	$x + 1$

(b) Multiplication

Reading the GF(2³) Tables

We can now read additions and multiplications from the table easily. Here are detailed examples:

Addition Example

Consider binary $100 + 010 = 110$.

This is equivalent to $x^2 + x$ in polynomial notation.

Verification: $(x^2) + (x) = x^2 + x$ (no reduction needed as degree < 3)

Multiplication Example

Consider binary $100 \times 010 = 011$.

In polynomial notation: $x^2 \times x = x^3$

Now reduce modulo $(x^3 + x + 1)$:

- $x^3 \bmod (x^3 + x + 1) = x + 1$
- This is because $x^3 = (x^3 + x + 1) - (x + 1)$, so $x^3 \equiv -(x+1) \equiv x+1 \pmod{2}$
- $x + 1$ is equivalent to 011 in binary

Key Insight

The reduction step is what makes the arithmetic work in a finite field. Without it, multiplication could produce polynomials of arbitrarily high degree.

Finding Multiplicative Inverse in $GF(2^n)$

Extended Euclidean Algorithm Adaptation

Just as the Euclidean algorithm can be adapted to find the greatest common divisor of two polynomials, the extended Euclidean algorithm can be adapted to find the multiplicative inverse of a polynomial.

Conditions for Finding Inverses

Specifically, the algorithm will find the multiplicative inverse of $b(x)$ modulo $a(x)$ if the degree of $b(x)$ is less than the degree of $a(x)$ and $\gcd[a(x), b(x)] = 1$.

Irreducible Polynomials

If $a(x)$ is an irreducible polynomial, then it has no factor other than itself or 1, so that $\gcd[a(x), b(x)] = 1$. This ensures that every nonzero polynomial has a multiplicative inverse.

The Equation to Solve

Given polynomials $a(x)$ and $b(x)$ with the degree of $a(x)$ greater than the degree of $b(x)$, we wish to solve the following equation for the values $v(x)$, $w(x)$, and $d(x)$, where $d(x) = \gcd[a(x), b(x)]$:

$$a(x)v(x) + b(x)w(x) = d(x)$$

If $d(x) = 1$, then $w(x)$ is the multiplicative inverse of $b(x)$ modulo $a(x)$.

Extended Euclidean Algorithm for Polynomials

Extended Euclidean Algorithm for Polynomials			
Calculate	Which satisfies	Calculate	Which satisfies
$r_{-1}(x) = a(x)$		$v_{-1}(x) = 1; w_{-1}(x) = 0$	$a(x) = a(x)v_{-1}(x) + bw_{-1}(x)$
$r_0(x) = b(x)$		$v_0(x) = 0; w_0(x) = 1$	$b(x) = a(x)v_0(x) + b(x)w_0(x)$
$r_1(x) = a(x) \bmod b(x)$ $q_1(x) = \text{quotient of } a(x)/b(x)$	$a(x) = q_1(x)b(x) + r_1(x)$	$v_1(x) = v_{-1}(x) - q_1(x)v_0(x) = 1$ $w_1(x) = w_{-1}(x) - q_1(x)w_0(x) = -q_1(x)$	$r_1(x) = a(x)v_1(x) + b(x)w_1(x)$
$r_2(x) = b(x) \bmod r_1(x)$ $q_2(x) = \text{quotient of } b(x)/r_1(x)$	$b(x) = q_2(x)r_1(x) + r_2(x)$	$v_2(x) = v_0(x) - q_2(x)v_1(x)$ $w_2(x) = w_0(x) - q_2(x)w_1(x)$	$r_2(x) = a(x)v_2(x) + b(x)w_2(x)$
$r_3(x) = r_1(x) \bmod r_2(x)$ $q_3(x) = \text{quotient of } r_1(x)/r_2(x)$	$r_1(x) = q_3(x)r_2(x) + r_3(x)$	$v_3(x) = v_1(x) - q_3(x)v_2(x)$ $w_3(x) = w_1(x) - q_3(x)w_2(x)$	$r_3(x) = a(x)v_3(x) + b(x)w_3(x)$
• • •	• • •	• • ⋮	• • •
$r_n(x) = r_{n-2}(x) \bmod r_{n-1}(x)$ $q_n(x) = \text{quotient of } r_{n-2}(x)/r_{n-1}(x)$	$r_{n-2}(x) = q_n(x)r_{n-1}(x) + r_n(x)$	$v_n(x) = v_{n-2}(x) - q_n(x)v_{n-1}(x)$ $w_n(x) = w_{n-2}(x) - q_n(x)w_{n-1}(x)$	$r_n(x) = a(x)v_n(x) + b(x)w_n(x)$
$r_{n+1}(x) = r_{n-1}(x) \bmod r_n(x) = 0$ $q_{n+1}(x) = \text{quotient of } r_{n-1}(x)/r_n(x)$	$r_{n-1}(x) = q_{n+1}(x)r_n(x) + 0$		$d(x) = \gcd(a(x), b(x)) = r_n(x)$ $v(x) = v_n(x); w(x) = w_n(x)$

Table 5.4: Extended Euclid Example

Initialization	$a(x) = x^8 + x^4 + x^3 + x + 1; v_{-1}(x) = 1; w_{-1}(x) = 0$ $b(x) = x^7 + x + 1; v_0(x) = 0; w_0(x) = 1$
Iteration 1	$q_1(x) = x; r_1(x) = x^4 + x^3 + x^2 + 1$ $v_1(x) = 1; w_1(x) = x$
Iteration 2	$q_2(x) = x^3 + x^2 + 1; r_2(x) = x$ $v_2(x) = x^3 + x^2 + 1; w_2(x) = x^4 + x^3 + x + 1$
Iteration 3	$q_3(x) = x^3 + x^2 + x; r_3(x) = 1$ $v_3(x) = x^6 + x^2 + x + 1; w_3(x) = x^7$
Iteration 4	$q_4(x) = x; r_4(x) = 0$ $v_4(x) = x^7 + x + 1; w_4(x) = x^8 + x^4 + x^3 + x + 1$
Result	$d(x) = r_3(x) = \gcd(a(x), b(x)) = 1$ $w(x) = w_3(x) = (x^7 + x + 1)^{-1} \bmod (x^8 + x^4 + x^3 + x + 1) = x^7$

Computational Considerations: Representation

Polynomial Representation

A polynomial $f(x)$ in $GF(2^n)$ of the form:

$$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0 = \sum_{i=0}^{n-1} a_i x^i$$

can be uniquely represented by the sequence of its n binary coefficients $(a_{n-1}, a_{n-2}, \dots, a_0)$.

Binary Representation

Thus, every polynomial in $GF(2^n)$ can be represented by an n -bit number. This makes $GF(2^n)$ arithmetic particularly efficient for computer implementation.

Tables 5.2 and 5.3

Tables 5.2 and 5.3 show the addition and multiplication tables for $GF(2^3)$ modulo $m(x) = (x^3 + x + 1)$. Table 5.2 uses the binary representation, and Table 5.3 uses the polynomial representation.

Advantage: The binary representation allows efficient hardware and software implementation using bitwise operations.

Addition in $GF(2^n)$

Addition as XOR Operation

We have seen that addition of polynomials is performed by adding corresponding coefficients, and, in the case of polynomials over Z_2 , addition is just the XOR operation. So, addition of two polynomials in $GF(2^n)$ corresponds to a bitwise XOR operation.

Example in $GF(2^8)$

Consider the two polynomials in $GF(2^8)$ from our earlier example:

- $f(x) = x^6 + x^4 + x^2 + x + 1$
- $g(x) = x^7 + x + 1$

Calculation

Polynomial notation: $(x^6 + x^4 + x^2 + x + 1) + (x^7 + x + 1) = x^7 + x^6 + x^4 + x^2$

Binary notation: $(01010111) \oplus (10000011) = (11010100)$

Hexadecimal notation: $\{57\} \oplus \{83\} = \{D4\}$

Implementation: Addition in $GF(2^n)$ is extremely efficient - it requires only n XOR operations, which are among the fastest operations in digital logic.

Multiplication in $GF(2^n)$

No Simple XOR for Multiplication

There is no simple XOR operation that will accomplish multiplication in $GF(2^n)$. However, a reasonably straightforward, easily implemented technique is available.

The Technique

We will discuss the technique with reference to $GF(2^8)$ using $m(x) = x^8 + x^4 + x^3 + x + 1$, which is the finite field used in AES. The technique readily generalizes to $GF(2^n)$.

Key Observation

The technique is based on the observation that:

$$x^8 \bmod m(x) = [m(x) - x^8] = (x^4 + x^3 + x + 1)$$

General Formula

In general, in $GF(2^n)$ with an n th-degree polynomial $p(x)$, we have:

$$x^n \bmod p(x) = [p(x) - x^n]$$

Multiplication by x in GF(2⁸)

Multiplication by x

Consider a polynomial in GF(2⁸), which has the form $f(x) = b_7x^7 + b_6x^6 + b_5x^5 + b_4x^4 + b_3x^3 + b_2x^2 + b_1x + b_0$. If we multiply by x:

$$x \times f(x) = (b_7x^8 + b_6x^7 + b_5x^6 + b_4x^5 + b_3x^4 + b_2x^3 + b_1x^2 + b_0x) \bmod m(x)$$

Two Cases

Case 1: $b_7 = 0$

If $b_7 = 0$, then the result is a polynomial of degree less than 8, which is already in reduced form, and no further computation is necessary.

Case 2: $b_7 = 1$

If $b_7 = 1$, then reduction modulo $m(x)$ is achieved using the formula from the previous slide.

Implementation Formula

$x \times f(x) =$

- $(b_6b_5b_4b_3b_2b_1b_00)$ if $b_7 = 0$
- $(b_6b_5b_4b_3b_2b_1b_00) \oplus (00011011)$ if $b_7 = 1$

It follows that multiplication by x (i.e., 00000010) can be implemented as a 1-bit left shift followed by a conditional bitwise XOR with (00011011), which represents $(x^4 + x^3 + x + 1)$.

Multiplication Example in GF(2⁸)

Problem

In an earlier example, we showed that for $f(x) = x^6 + x^4 + x^2 + x + 1$, $g(x) = x^7 + x + 1$, and $m(x) = x^8 + x^4 + x^3 + x + 1$, we have $f(x) \times g(x) \bmod m(x) = x^7 + x^6 + 1$.

Binary Calculation

Redoing this in binary arithmetic, we need to compute $(01010111) \times (10000011)$. First, we determine the results of multiplication by powers of x :

$$(01010111) \times (00000010) = (10101110)$$

$$(01010111) \times (00000100) = (01011100) \oplus (00011011) = (01000111)$$

$$(01010111) \times (00001000) = (10001110)$$

$$(01010111) \times (00010000) = (00011100) \oplus (00011011) = (00000111)$$

$$(01010111) \times (00100000) = (00001110)$$

$$(01010111) \times (01000000) = (00011100)$$

$$(01010111) \times (10000000) = (00111000)$$

Final Calculation

$$\begin{aligned}(01010111) \times (10000011) &= (01010111) \times [(00000001) \oplus (00000010) \oplus (10000000)] \\ &= (01010111) \oplus (10101110) \oplus (00111000) = \mathbf{(11000001)}\end{aligned}$$

which is equivalent to $x^7 + x^6 + 1$.



Acknowledgement

The material presented in these slides incorporates content from books, academic lecture notes, online resources, and generated text. All rights remain with the original authors and publishers. This content is used solely for educational purposes to illustrate and clarify concepts. No copyright infringement is intended.